Iftikhar Ahmed

Research Assistant, Big-Matrix R&D Lab

Seeking Master's and PhD positions

ifti3061@gmail.com +880 1533 133309 Dhaka, Bangladesh

Google Scholar research.iftikharahmed.site NinjaIfti iftikhar-ahmed

Research Interests

Trustworthy machine learning: models that stay reliable under adversarial attack, distribution shift and limited measurement. I work on three threads: **robust and cross-lingual audio deepfake detection**; **AI for radiotherapy and oncology** (target delineation, medical image segmentation, multimodal outcome prediction); and **verifiable quantum machine learning**.

Publications

[C1] **Iftikhar Ahmed**, Mueid Islam Arian, Nusrat Jahan Lisa. "Mel-Spectrogram Representations for CNN Based Adversarial Defense for Audio Deepfake Detection." In *Proceedings of the 11th International Conference on Machine Learning Technologies (ICMLT 2026)*, Germany. IEEE, 2026.

IEEE Xplore DOI: 10.1109/ICMLT69916.2026.11689539

> Mel-spectrograms beat LFCC for CNN detection on WaveFake (EER 0.034 vs 0.053, a 36% improvement) across leave-one-vocoder-out validation on seven neural vocoders. Adaptive adversarial training lowers EER under FGSM from 0.70 to 0.107, and under PGD from 0.90 to 0.154.

[S1] **Iftikhar Ahmed**, M. N. Mithila, A. Dhrubo, M. A. Qayum, M. Khan. "DASF-Net: Dual-Adversarial Spectral Fusion for Cross-Lingual and Adversarially Robust Audio Deepfake Detection." *Under review*, 2026. Preprint

Ongoing Research

Cross-Lingual Audio Deepfake Detection: English and Bengali — In preparation
First author · Speech security

Clinical Target Volume Delineation for Radiotherapy — Ongoing
Big-Matrix R&D Lab · Supervisor: Prof. Mohammad Monir Uddin

Multimodal AI for Treatment Response and Outcome Prediction in Lung Cancer — Proposed
Big-Matrix R&D Lab · PI: Prof. Mohammad Monir Uddin · With a partner cancer hospital

Foundation-Model Segmentation for Breast Radiotherapy CT — Ongoing
Medical image segmentation

Region Selection in Whole-Slide Pathology Images — Ongoing
Computational pathology · Quantum optimisation

ZK-QGuard: Auditable Inference for Private Quantum Cloud Services — Team lead
Leading a five-person team · Quantum ML · Zero-knowledge proofs

Undergraduate Thesis

LectureAssist: An Agentic Multimodal AI Assistant that Turns Lectures into Adaptive Study Material 2026
North South University · Supervisor: Dr. Md Adnan Arefeen · With Salman Shafi and Jarinah Tasnim

Scene-adaptive OCR and Whisper transcription feed a multi-agent pipeline that plans, generates, validates and refines five question types, served by Gemma 3 on a single consumer GPU, all locally hosted. Agentic and one-shot generation are compared with an LLM-as-judge evaluation.

Completed Research Projects

Polyp Segmentation in Colonoscopy Images — Code
Biomedical image segmentation

U-Net with an ImageNet-pretrained ResNet34 encoder on Kvasir-SEG, trained with BCE + Dice loss. Mean test Dice 0.910 and IoU 0.855 at original resolution, with a failure analysis on pedunculated and flat polyps.

Emotion Recognition from Hand-Sign Gestures
Computer vision

A computer vision system that identifies a user's emotion from hand-sign gestures.

Safety Equipment Detection CODE

Object detection

A YOLOv5 object detector for safety equipment.

Education

North South University Dhaka, Bangladesh

B.Sc. in Computer Science and Engineering · CGPA 3.07/4.00 2021 – 2026

Research Experience

Research Assistant Big-Matrix R&D Lab

AI for healthcare: radiotherapy target delineation, oncology outcome prediction Present

Technical Skills

ML & research: PyTorch, scikit-learn, NumPy, Hugging Face, nnU-Net, TotalSegmentator, DINOv3, Whisper, adversarial training (FGSM, PGD), multiple-instance learning

Quantum: Qiskit, QUBO formulation, simulated annealing

Languages & tools: Python, LaTeX, Git, Kaggle and Colab GPU workflows

References

Dr. Nusrat Jahan Lisa
Assistant Professor, Dept. of ECE
North South University
nusrat.lisa@northsouth.edu

Ahasan Habib Nirjon
Data Architect Specialist
Banglalink Digital Communications Ltd.
ahasan.nirjon@banglalink.net